1.3 How robots work

In this lesson

You will learn:

- how a robot uses sensors, a controller and actuators to do its work.

What makes a robot work?

You must balance a tennis ball on a book and walk across the room without dropping the ball. How do you do this?

As you cross the room, you use your senses to warn you when the ball is starting to roll. Your brain works out how to stop the ball from falling. Your brain tells your muscles how to move your arm to stop the ball rolling.

A robot works in the same way.

- A robot has **sensors** that act like your eyes and ears. They tell the robot what is happening in the world around it.
- A **controller** is the brain of a robot.
- **Actuators** are the muscles of a robot. They make the robot move.

Sensors

A robot does not have eyes and ears like humans. It uses components called sensors.

- Video cameras give the robot sight.
- **Proximity sensors** tell a robot how close an object is.
- **Bumpers** tell the robot it has collided with something.
- **Pressure sensors** let a robot know that it is touching something.
- Sensors that detect chemicals give a robot its sense of smell and taste.

A robot uses the sensors it needs to do its job.



Controller

A controller is the brain of a robot. The controller is programmed to do a job. A robot cannot think like a human. It carries out whatever instructions it has been programmed to do.

Actuators

A robot does not have muscles and bones like humans. It uses mechanical parts to lift and move objects. The mechanical parts that make a robot move are called actuators. The actuators in a robot do the same job as the muscles in your arm.

How the robot vacuum cleaner works

A robot vacuum cleaner helps to clean a house. A motor is the only extra mechanical part that a robot vacuum cleaner has compared to a normal cleaner. The motor moves the cleaner around the home on wheels.

Proximity sensors measure a room so that the cleaner can calculate the best route to take. Proximity sensors use a light beam to calculate distance. When the light beam hits something solid, it bounces back to the sensor. The time it takes to bounce back tells the sensor how far the light travelled.



A sensor called a bumper detects if the cleaner hits a wall or piece of furniture. If the cleaner detects a collision, it follows an algorithm. The cleaner makes a small turn and tries to move again. Once it finds a clear path it continues cleaning.

The cleaner has a **vibration sensor** on the base. If a lot of dirt hits the sensor, the vibration it causes tells the cleaner that area needs more cleaning.

A sensor warns the cleaner if the battery power is low. The cleaner finds an electrical charging unit using proximity sensors. The cleaner recharges its batteries before continuing cleaning.



Activity

Give two examples of the sensors a robot vacuum cleaner uses. How are the sensors used?

Think again

What is an actuator? Why does a robot need actuators?



Extra challenge

Have you seen a 'barista' in a coffee shop? A barista is the person who makes the cups of coffee. Could a robot be made that works as a barista? Describe how sensors and actuators might be used by a robot barista.

1.4 What is a control system?

In this lesson

You will learn:

- what control systems are used for
- what a control loop does.



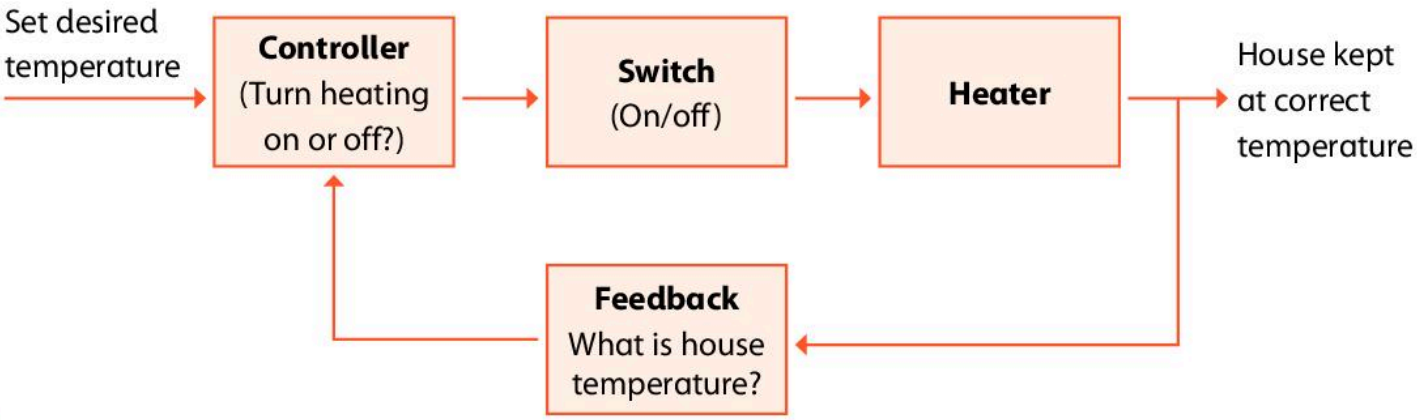
What is a control loop?

If you have a heating system in your home, you may have seen a controller like this:

A heating controller lets you set an ideal temperature for your home. Suppose your ideal temperature is 20 degrees Celsius.

- When the heating is switched on, the controller sends a message to turn on the radiators.
- The message passes through a **temperature sensor**.
- The sensor sends a message back to the controller telling it the actual temperature in the house.
- When the temperature reaches 20 degrees, the controller turns off the radiators.
- The temperature sensor keeps sending messages back to the controller. When the temperature falls below 20 degrees, the radiators are turned back on.

This is a **control loop**. It is sometimes called a **feedback loop**. The sensor feeds back information to the controller.



Control systems in the home



Control systems are installed in many homes. They control temperature, lighting and security.

Modern homes may have many sensors installed:

- Heating systems use temperature sensors.
- Alarm systems use proximity sensors and pressure pads to detect intruders.
- Security systems use video cameras.
- Lighting systems use proximity sensors to detect movement.

Lights can be turned on automatically when a sensor detects someone entering a room. They are turned off again when the sensor detects the room is unused. Heating can be kept at a lower level when the house is empty and warmer when people are in it.

The controller is connected to the internet, so the house owner can control systems from anywhere. Heating can be turned on so the house is warm when people arrive home.

Devices such as ovens can be turned on so food is cooked when people arrive home from work. Messages are sent to the home owner if an intruder is detected. Security cameras can be viewed on a smartphone. The home owner can even view the contents of their fridge to find out what they need to buy from the supermarket.

A house with control systems is called a **smart home**. Control systems make life easier, more secure and reduce energy use.



Activity

You are travelling home in your robot car with your parents. What messages can you send to your smart home so that it is welcoming when you arrive?

Think again

A feedback loop can be used to control air conditioning to keep a home at a low temperature. Draw the feedback loop (or control loop) for this process.



Extra challenge

Use your program design skills to write an algorithm. The algorithm should describe how a heating controller keeps the temperature in the house at 18 degrees Celsius.